

# Stuyvesant Faculty Perspectives: Ms. Jessica Novillo

## Unveiling the Passion - A Journey into Computer Science Education

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*In the rapidly advancing field of technological advancement, educators serve as the guiding lights, shaping the minds of tomorrow's innovators. Through an insightful interview with Ms. Novillo, a distinguished software developer turned fervent computer science educator at Stuyvesant High School, we examine of her transition from the realm of coding to the realm of teaching, exploring her motivations, aspirations, and the profound impact she seeks to impart upon her students.*

*Ms. Novillo's answers are taken from an interview conducted by Aditya Anand and have been edited into a narrative structure.*

### 1 Getting Started

Ms. Novillo's trajectory into the realm of computer science traces back to her formative years in high school. Under the mentorship of an inspiring instructor, she discovered her passion for coding and embarked on a journey of exploration and discovery as she learned topics such as software development. This initial spark kindled a lasting interest in the field, prompting her to defy familial expectations and pursue a career path aligned with her passions.

With a robust background spanning fourteen years in software development, Ms. Novillo embarked on a new chapter in her professional journey as an educator. Her transition to teaching marked the realization of a long-cherished dream, as she found herself at Stuyvesant High School, where she could impart knowledge and foster intellectual curiosity among her students. Offering courses ranging from AP Computer Science to software development, Ms. Novillo endeavors to equip her pupils with not only technical proficiency but also essential skills such as creativity, critical thinking, and problem-solving. She accomplishes this through an interactive lesson-based class. Usually starting with a presentation about the day's topic, she helps students gain an understanding of the material by allowing them to talk in pairs and use whiteboards to display trace diagrams. The remaining portion of the period, ranging from half to three-quarters of the period, is spent working on code for labs and classwork.

### 2 Looking Back

Reflecting on her inaugural year at Stuyvesant, Ms. Novillo exudes enthusiasm and gratitude for the enriching experience she has encountered. Immersed in a supportive environment with dedicated colleagues and eager learners, she perceives every challenge as an opportunity for personal and professional growth. Despite the constraints posed by time, her unwavering commitment to inspiring and empowering her students remains steadfast.

When contemplating the rewards and challenges of teaching, Ms. Novillo's passion for her vocation becomes palpable. She derives immense fulfillment from witnessing the intellectual growth and achievements of her students, cherishing each moment spent within the classroom. However, she acknowledges the demanding nature of her profession, where time constraints present obstacles yet serve as catalysts for continual improvement and innovation. Between the commute to Stuyvesant and the grading of labs and tests in late hours, the job can limit one's ability to spend time on their own hobbies.

Ms. Novillo places a strong emphasis on the value of research in the field of computer science education. Recognizing that not everything can be taught within the confines of a single class period, she encourages her students to delve deeper into topics of interest, fostering a culture of curiosity and exploration. Through diligent research and self-directed learning, students can augment their understanding of complex concepts and stay abreast of advancements in the ever-evolving landscape of technology.

### **3 Alternate Paths**

In a poignant revelation, Ms. Novillo shares her affinity for mathematics, particularly calculus, highlighting the breadth of her academic interests. Despite her steadfast dedication to computer science, she acknowledges the allure of mathematics and envisions a parallel universe where she imparts the intricacies of calculus to eager minds. As she remained in the STEM path, Ms. Novillo's decision aids many students today. Ultimately, Ms. Novillo extends a heartfelt invitation to students, beckoning them to embark on a transformative journey through the realms of computer science education. In her classroom, learning transcends the confines of traditional pedagogy, evolving into a collaborative odyssey fueled by curiosity, innovation, and the joy of discovery. Ms. Novillo's words resonate, echoing the sentiment that computer science, as a beacon of hope, illuminates the path to a brighter, more empowered future.